Research Question 1: After accounting for sampling variability, and data sparsity, what are the robust, latent estimates of age-adjusted asthma hospitalization rates across U.S. states? Which states exhibit a significantly elevated burden compared to the national average?

Data Overview:

The dataset that we are basing our findings from are based on a release from the U.S. Chronic Disease Indicators(CDI) dataset published by the Center for Disease Control and Prevention(CDC) released the year of 2023. Findings are available publicly through the CDC data portal. It is also mirrored on Kaggle's public domain, allowing anyone to both use and modify studies on the data. The data source we used came from Kaggle, which contains indicators for numerous chronic conditions, mixed stratification groups, non-numeric entries and metadata footnotes embedded within the data columns. Before extracting a consistent analytic subset, we require a substantial preprocessing for the asthma related hospitalization rates.

Each row in our analytic dataset represents a state, year, and age-adjusted asthma hospitalization rate cases. Therefore, the granularity is an aggregated epidemiological indicator. This aggregation does not allow us to analyze fully within state variations. It allowed for our findings only to be on a statewide trend, rather than individual or hospital level pattern trends. Because datasets from the CDI do not include at-risk populace or raw hospitalization counts, our sampling variability is not a direct computation and cannot use a count-based likelihood. Instead, we were able to use confidence intervals, which were reported by the CDC at 95% which approximated the measurement-error standard deviation for each state-year observation, which also allowed us to incorporate the uncertainty directly into our hierarchical normal likelihood. Measurement error is seen present mainly due to hospitalization rate estimates varying in precision, making several data quality concerns relevant. This allows for selection bias to arise not only with subset reports of asthma hospitalization indicators by state, but also by states which carry more complete data reports by being larger and better resourced. Smaller states having a much wider confidence interval can lead to unstable observed rates due to the report practices varying by state and even time. Convenience sampling is inherent to the dataset which shows the importance of modeling heterogeneous uncertainty by emphasizing these issues including states with missing data.

Despite these findings, the dataset does not include raw hospitalization counts or even popular denominators. We also lack demographic information that would serve to further understand if asthma burdens are due to those certain population structures such as age, race, and gender. These would be important columns that would serve to be useful factors by modeling the

use of Poisson or Binomial likelihoods from the underlying data but are unavailable. These features would make the model have more interpretable comparisons across the states.

Despite all of this, we are presented with missing data in our findings. Many of the observations by state-year appear as bare or “data not available”. This is due to the underlying sample size being too small for the CDC to publish these reliable rates. Having this repetitive pattern reflect, especially on small population states, shows the reason why only 34 total states have appeared across the entire period for data findings. Because of the missing information being systematic rather than random, it would start creating bias on the data. Instead of having that, we exclude the reported values and heavily rely on a model of hierarchy to stabilize estimates for states with low, or sparse reporting. This then accounts for differing levels of uncertainty across the states to be partially pooled. We then filtered the dataset to retain only the observations relevant to our study, the topic: “Asthma”, the indicator: “Hospitalizations for Asthma”, the age-adjusted rate, and the overall stratification category. Age-adjusted hospitalization rates were by their ability to best reflect underlying asthma burden. Being the only stable and comparable measure across the states, this makes it a comparable measure across the states with different age distributions. These decisions allowed for our models to naturally account for imbalances in the dataset and to realistically propagate uncertainty into the partial pooling process.

Research Question:

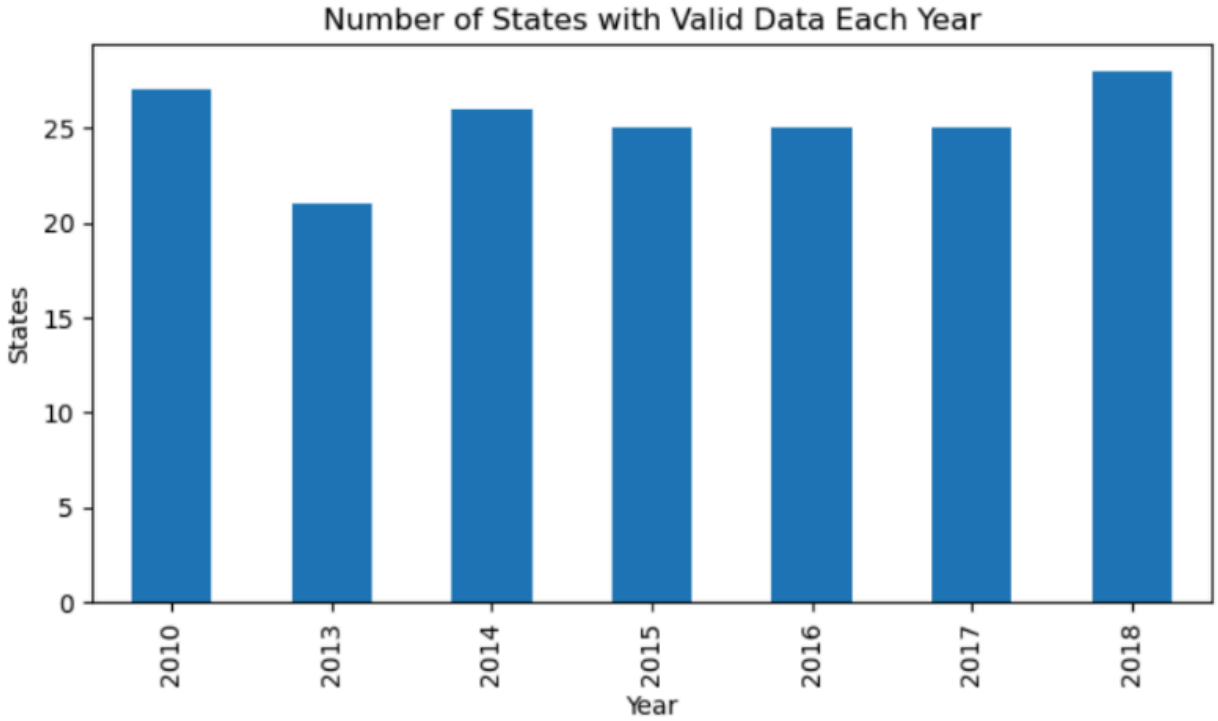
Our research question is how do asthma hospitalization rates vary across the U.S? Understanding the differences across different states can help public health agencies identify where asthma control and additional resources might be needed. We are using a Bayesian hierarchical model with particle pooling suited to this question. Due to the unbalanced and sparse dataset, partial pooling allows states with limited data to borrow strength from the overall distribution to produce more stable and comparable estimates. And the bayesian framework will let us incorporate CDC reported confidence intervals directly as measurement error. The hierarchical Normal likelihood allows us to appropriately weight each observation according to its precision, ensuring that noisy estimates do not disproportionately influence the results. These modeling features directly support the goal of estimating underlying state-level hospitalization burdens.

The model assumes states are partially exchangeable and that the reported rates are approximately normally distributed. Which these may not always hold. If some states differ in systematic ways such as environmental exposures. Then the model would be overshrink on the meaningful differences. The missing data also is not random, since many small states do not report hospitalization rates, it will limit how generalizable our findings are.

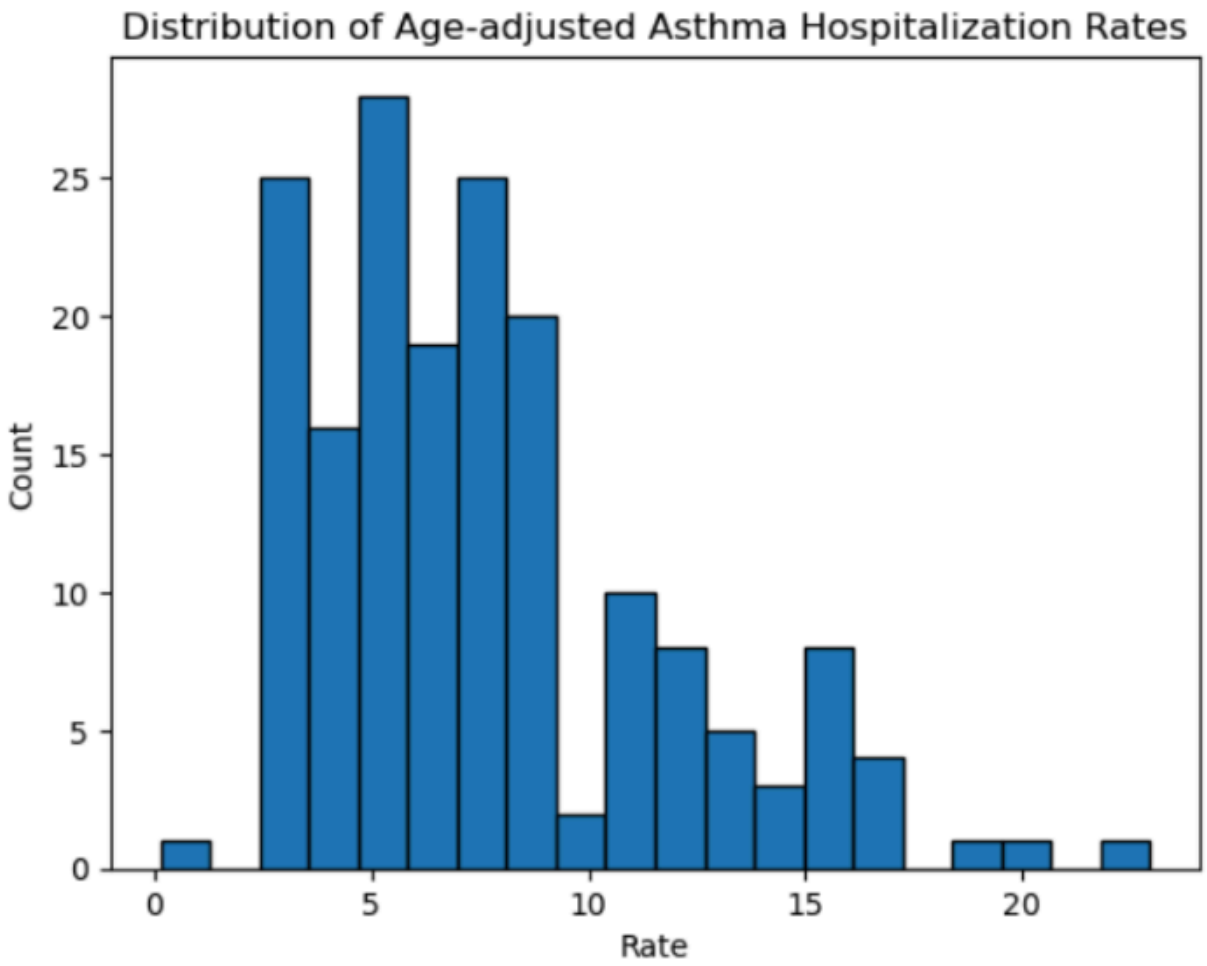
Prior Work:

Camargo et al. (2024) had quantified cases of persistent, severe asthma across the U.S. states which expressed burdens for cases relating to asthma. Based on the highlights of their studies which stated health outcomes widely differ across the states, it further supported our research question, regarding hospitalization rates. Having our hierarchical model able to obtain stable estimated state levels, it further shows how their clinical data aligns closely with our motivation. The study shows why geographic comparison of asthma is important and hierarchical modeling is appropriate when data are noisy. Our approach is simpler but follows the same logic. We aim to produce stable, comparable state-level estimates despite missingness and measurement error in the CDI data.

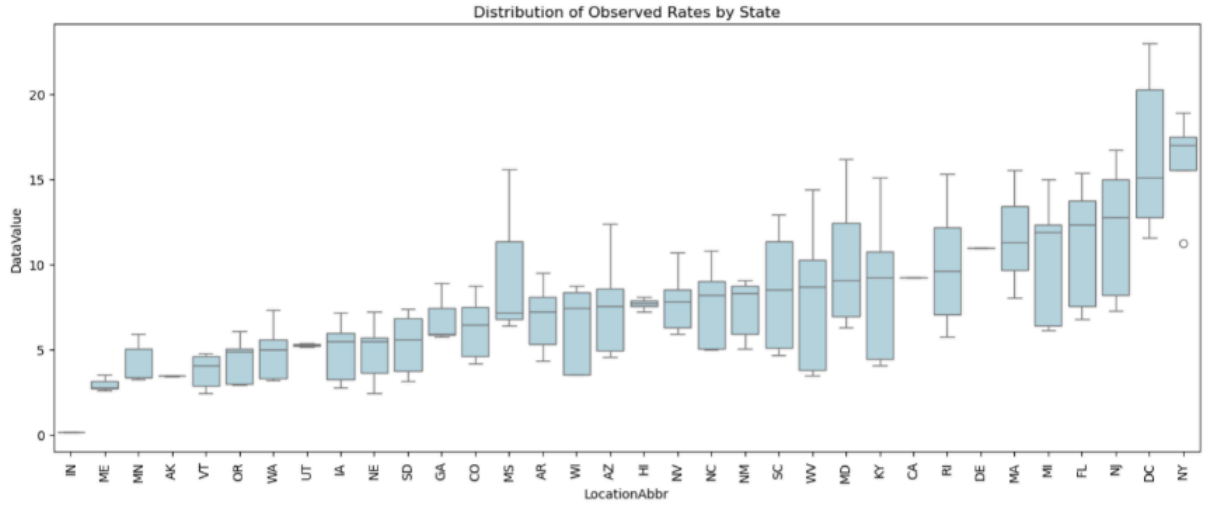
EDA:



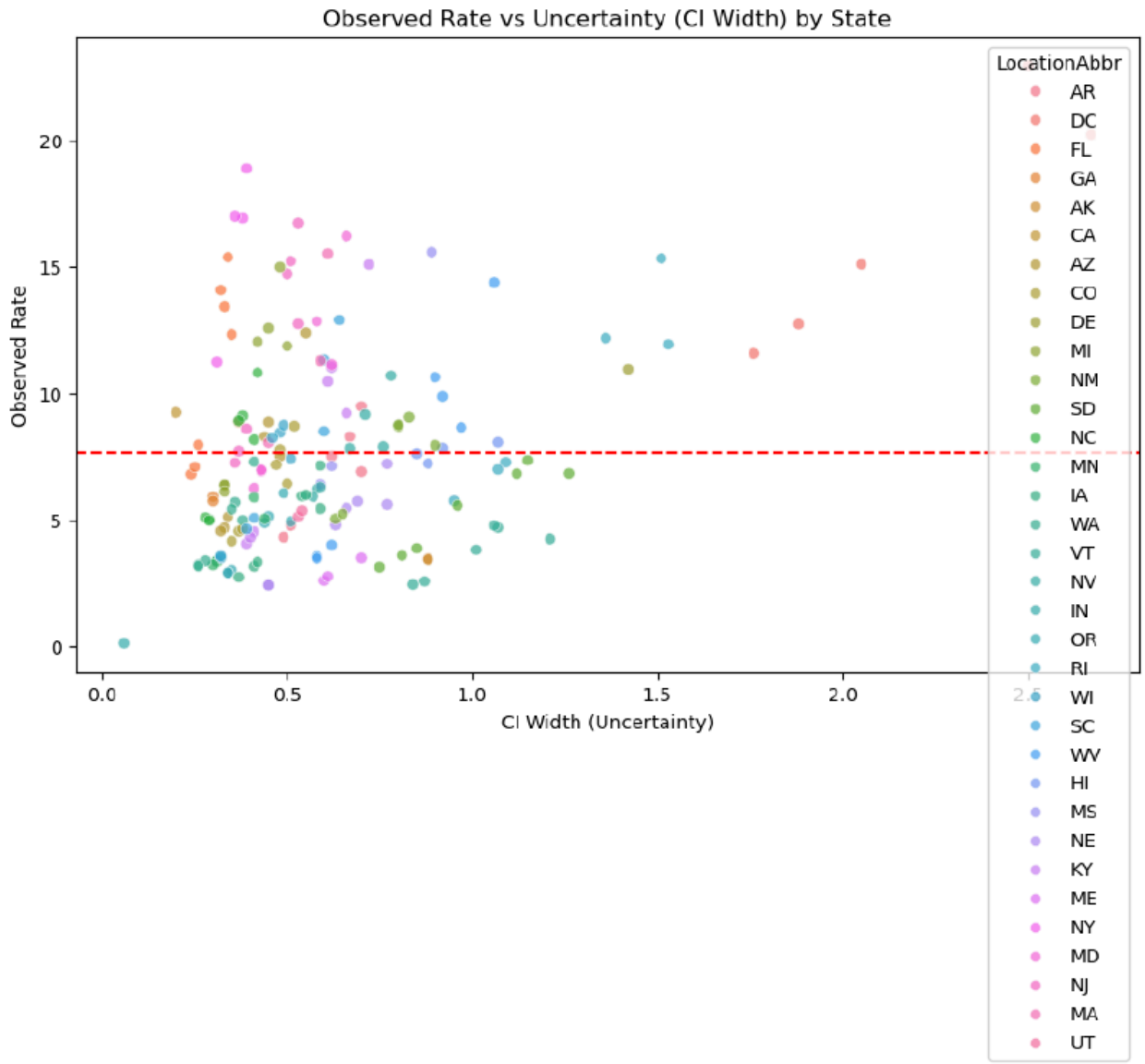
This plot shows that we only have asthma hospitalization rates for a total of 21-28 states in any given year. There is not a single year where we have complete information for all states. This is due to the dataset being sparse and unbalanced. These findings led us to use a hierarchical model. This'd allow us to produce stable estimates and merge information across states when data is limited.



This histogram shows that hospitalizations are continuous, unimodal, and generally symmetric distributions. Most values lie in the 4-10 per 10,000 residents. A number of high values present a slight right tail, but overall, the distribution curve provides evidence that a Normal likelihood is a reasonable modeling choice. This supports treating the observed rates as noisy estimates of underlying state-level hospitalization.



This boxplot tells there is a variability between the states and the within state fluctuation differs from year to year. Some states show stable hospitalization patterns and others show large swings. This can be interpreted that sampling noise can drive the extreme state values with less observations.These patterns justify the partial pooling. They shrink noisy state-level estimates toward the population mean.



This scatterplot shows the states with the largest confidence intervals are often the small-population states such as DC, DE, RI, VT, and SD. These populations usually have the more extreme observed rates. The high or low indicators from the national average reflect measurement error. Rather having reflected measurement errors, instead of true differences, this demonstrates the connection between CI width and deviation from our national average directly. This is through uncertain measurements that are incorporated by the hierarchical model.

From all four visualizations above, we observe that we have three key patterns to the finding. The first is that data availability fluctuates across years and states. The second is that the hospitalization rates vary substantially from state to state. The third is to have pretty wide variation in measurement uncertainty varies, while smaller states also will have relatively larger CI width. Overall, these findings indicate that raw observed rates are heavily affected by sampling noise. A Bayesian hierarchical model helps to tackle the challenges mentioned before. It enables us to borrow information across states and accounting for measurement error. Therefore we obtain more valid and stable state-level estimates.

RQS (Methods, Results, Discussion):

<u>Methods:</u>

To estimate the underlying asthma hospitalization burden across states. We use a Bayesian hierarchical Normal–Normal model that separates sampling noise from true geographic differences. Each observed hospitalization rate y_i is treated as a noisy estimate of the latent rate $\theta_{s[i]}$ for state $s[i]$. Because the CDI dataset does not provide raw hospitalization counts and population denominators. We decided to approximate each observation's standard error using the CDC-reported 95% confidence interval:

$$\sigma_i = \frac{\text{UpperCI}_i - \text{LowerCI}_i}{2 \cdot 1.96}$$

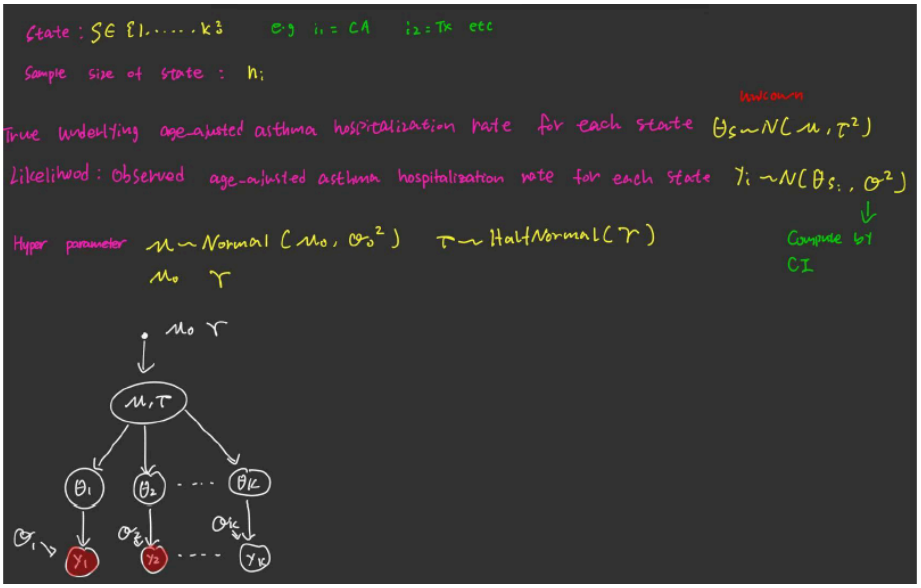
Because our goal is to estimate a stable, state-level burden. We treat state-year observations as noisy measurements of a single underlying θ_state. The observation model is therefore:

$$y_i \sim N(\theta_{s[i]}, \sigma_i^2)$$

We assume true hospitalization rates follow a partially pooled distribution at the state-level:

$$\theta_s \sim N(\mu, \tau^2)$$

When μ represents the national average and τ captures between states variability. We place weakly informative hyperpriors: $\mu \sim N(7, 5^2), \quad \tau \sim \text{HalfNormal}(5)$ chosen based on the empirical range of observed rates. For the posterior inference we performed inNUTS sampling.



This hierarchical framework is appropriate because the dataset is sparse and unbalanced, with only 21–28 states reporting in any given year and large differences in measurement uncertainty across states. Partial pooling stabilizes estimates for small-population states with wide confidence intervals while preserving meaningful deviations for states with precise data. The model also naturally incorporates heterogeneous uncertainty through sigma to ensure that noisier observations receive less weight.

Several assumptions underlie this approach. The model assumes that observed age-adjusted rates are approximately Normally distributed and that states are partially exchangeable. It also assumes that CDC confidence intervals accurately reflect sampling variability. If some states differ in structural ways for example the environmental exposures. These assumptions may be violated. This will lead to biased estimates. Missing data are not random, as small-population states are more likely to have suppressed values. Therefore estimates for these states should be interpreted with caution. Despite these limitations, the hierarchical model provides a principled and interpretable approach for estimating state-level asthma within the constraints of the CDI dataset.

Results:

We selected weakly informative hyperpriors to reflect broad prior uncertainty while remaining consistent with our EDA. The national mean rate μ was assigned a $N(7, 5^2)$ prior because most observed rates fall between 4 and 10. The between-state standard deviation τ received a $\mathrm{HalfNormal}(5)$ prior, allowing substantial heterogeneity while avoiding unrealistic extremes. These choices do not impose strong assumptions and let the data primarily determine the posterior.

So we got posterior inference indicates a national mean asthma hospitalization rate of :

$$\mu \approx 6.98 \ (95\% \text{ HDI} : 5.82,\ 8.07)$$

The substantial between state standard deviation is:

$$\tau \approx 3.39 \ (95\% \text{ HDI} : 2.63,\ 4.22)$$

This indicates moderate-to-large heterogeneity across states. State-level posterior means ranged approximately from 0.13 (IN) to about 7.19 (SC). Small-sample states experienced noticeable shrinkage towards the national mean, which stabilizes noisy raw estimates. A counterintuitive finding is that several states appearing extreme in raw data were not extreme in the posterior. This demonstrates how sampling noise can distort observed hospitalization rates, reinforcing the benefit of partial pooling.

The sensitivity to hyperparameters. To evaluate robustness, we reduced the prior scale for μ and τ from $\gamma = 5$ to $\gamma = 3$. The maximum difference in any state-level posterior means was < 0.01, meaning that results are highly robust. This suggests the posterior is driven by the data. Despite the uneven uncertainty and not overly sensitive to prior choice.

Discussion:

Our hierarchical models have several limitations. It assumes that states are partially exchangeable and that CDC reported confidence intervals accurately represent sampling noise. In practice, states differ in environmental exposures, healthcare access, and demographics will have weaken in this assumption. This will lead to overshrink for states with genuinely extreme asthma burden. The small states are more likely to have suppressed values due to the missing data, so their estimates should be interpreted with caution. Future work could explore pattern mixture models to assess how sensitive our conclusions are to non-random missingness

Despite these limitations, the model converged and produced stable inference. More complex alternatives such as Poisson or Negative Binomial models were not feasible because the CDI dataset does not include raw hospitalization counts. Our findings align with our prior, which also reports substantial geographic variation in asthma burden across states. It shows that high rates in some regions persist even after accounting for noise. The main difference is that their analysis uses richer clinical dataset, while our results rely on partial pooling to stabilize noisy estimates from public health data. Both approaches highlight the geographic disparities in asthma remain meaningful and actionable.

We are missing important background information such as each state's air quality, poverty levels, and access to healthcare. Every signal of them is going to affect asthma hospitalization. If we had these, we can better understand which states truly have higher asthma risks instead of confusing those risks with other factors.

Conclusions:

The information we were able to gather regarding asthma effects vary differently between states and is something that was able to be presented through our hierarchical model. Larger states that are better prepared, and better equipped, present their data with a more stable stream plus, their asthma rates are closer to what we had originally observed. Whereas lower populace states can range between high or unusually high cases, just by chance. But, when we account for these uncertainties, then and only then are we able to clearly see the numbers sync with the national average. Having this has given us a more trustworthy, a more polished ideology of what states have actually increased asthma problems.

The analysis result is limited due to the lack of information in raw hospitalization counts, the inconsistent reporting across years, and larger variation in measurement uncertainty. If we were able to add additional domain knowledge into it, it could help us to clarify the geographic differences that reflect true disease burden. Our findings still are highly robust to changes in prior scale. Within the state-level estimates shifting by less than 0.01 under alternative hyperparameters. However, our findings are limited to state-level comparisons only and it should not be assumed for smaller geographic areas.

The future work should include richer covariates and finer version data to better explain state-level variation. This could also help to reduce reliance on exchangeability assumptions. Based on our findings, the states with high asthma hospitalization rates should get more help such as asthma helping programs.

Reference:

Camargo, Jr, Carlos A. “Home Page: Annals of Allergy, Asthma & Immunology.” *Annallergy*, May 2024, www.annallergy.org/article/S1081-1206(23)00561-6/fulltext.

Data C102 Group22 Research Q#2

Datasets Used (Data Overview Part I):

Units/Outcome Dataset (2011 - 2020):

U.S. Chronic Disease Indicators (CDI). *U.S. Chronic Disease Indicators (CDI) 2023 Release Dataset*, Centers for Disease Control and Prevention, 2023, https://www.kaggle.com/datasets/kadriyelidirim12/u-s-chronic-disease-indicators-cdi-2023-release.

Treatment Dataset (2011 - 2020):

Environmental Public Health Tracking (EPHT). *PM2.5: Highest Annual Average Concentration (Monitor + Modeled Data)*, Centers for Disease Control and Prevention, 2011–2020, https://ephtracking.cdc.gov/DataExplorer/.

Confounders Datasets (2011 - 2020):

(age-adjusted rate already coordinates the confounder effect from 'age')

Behavioral Risk Factor Surveillance System (BRFSS). *BRFSS Prevalence Data*, Centers for Disease Control and Prevention, 2011–2020, https://data.cdc.gov/Behavioral-Risk-Factors/Behavioral-Risk-Factor-Surveillance-System-BRFSS-P/dtwv-5yxu/data_preview.

American Community Survey (ACS). *ACS 1-Year Estimates: Median Household Income (Table S1901)*, U.S. Census Bureau, 2011–2020, https://data.census.gov/table/ACSST1Y2023.S1901.

Download Process:

U.S. Chronic Disease Indicators (CDI) 2023 Release Dataset and the BRFSS Prevalence Dataset were downloaded directly from the CDC open data portal using the built-in *Export* / *Download* options.

For the EPHT PM2.5 dataset, we first filtered the broader air quality database to PM2.5: Highest Annual Average Concentration (Monitor + Modeled Data, National by County) and then exported the filtered results for later aggregation to the state level.

ACS 1-Year Estimates for Median Household Income (Table S1901) and Age and Sex (Table S0101) were downloaded from data.census.gov by selecting each year through the *Year* filter and exporting the corresponding tables.

Analysis Plan:

Step#1: Data Overview Part II

Data Generation:

Our analysis combines multiple federal surveillance datasets that differ in how values are generated. Asthma hospitalizations (CDI) and PM2.5 exposure (EPHT) are population-level indicators derived from administrative health records and statewide modeled pollution estimates, while behavioral and socioeconomic confounders (smoking, obesity, physical inactivity, income) come from weighted probability samples in BRFSS and ACS. We harmonized variables across sources by aligning state–year identifiers, aggregating county-level PM2.5 to the state level, and merging all datasets into a unified state–year panel.

Sample-based datasets (BRFSS and ACS). These estimates are subject to sampling error and nonresponse, particularly in smaller states, but stratified sampling and survey weights support generalizability to state populations. As a result, remaining uncertainty primarily affects precision rather than introducing systematic bias.

Census-like administrative datasets (CDI and EPHT). Although not full population censuses, these systems are designed to represent statewide conditions using comprehensive hospital records and monitoring-plus-modeling approaches for PM2.5. While some measurement error is unavoidable, no major population groups are intentionally excluded.

Data Granularity:

Our dataset is structured at the state–year level, so each row corresponds to one state in a given year between 2011 and 2020. For every state–year pair, we combine that state's age-adjusted asthma hospitalization rate, its annual PM2.5 exposure, and a set of population-level behavioral and socioeconomic confounders such as smoking, obesity, and physical inactivity. Because these observations are aggregated at the state level, our findings speak to how changes in a state's average pollution levels relate to changes in its overall asthma hospitalization rates. In other words, we can infer causal relationships at the population scale, but not at the level of individual risk. This granularity still allows for meaningful causal analysis of environmental impacts on public health, as long as the results are interpreted as state-level effects rather than claims about individual behavior or susceptibility.

Data Quality:

Selection Bias:
Relevant for BRFSS confounders because survey participation is voluntary. We use weighting average to calculate prevalence of confounders, which reduces the bias.
Measurement Error:

Noise from modeled PM2.5, self-reported BRFSS behaviors, and hospital coding is unavoidable and likely attenuates effect sizes, but it does not invalidate a causal interpretation.

Convenience Sampling:

Not applicable to datasets used.

Unavailable Features Improvements:

Some potentially relevant variables are unavailable in our dataset. We lack individual-level hospitalization and behavioral data, limiting analysis of personal exposure pathways, as well as state-level healthcare system indicators (e.g., insurance coverage or care access) and policy changes that could influence hospitalization rates. While these omissions do not invalidate our analysis, they constrain our ability to examine underlying mechanisms and heterogeneity in effects.

Data Pre-processing Methods:

1. For Outcome Data (Hospitalization Rates):
   We standardized state names, removed territories, and ensured one observation per state–year. Missing outcome values were imputed using within-state linear interpolation,
   $$\hat{x}_{s,y} = x_{s,y_1} + \frac{y - y_1}{y_2 - y_1}(x_{s,y_2} - x_{s,y_1})$$
   and, when necessary, year-specific cross-state means,
   $$\hat{x}_{s,y} = \frac{1}{|S_y|} \sum_{s' \in S_y} x_{s',y}$$
2. For Confounders (Smoking, Obesity, Physical Inactivity):
   BRFSS data were filtered to relevant indicators and collapsed to one state–year value by combining male/female estimates via sample-size weighting,
   $$\hat{p}_{s,y} = \frac{n_m p_m + n_f p_f}{n_m + n_f}$$
   Any missing values were filled using the same interpolation and mean-imputation approach as outcomes. This ensured consistent, comparable confounder measures across states and years.
3. For Treatment (PM2.5 Exposure):
   County-level PM2.5_{2.5}2.5 measurements were cleaned and aggregated into state–year averages to match the granularity of other variables. This avoided geographic mismatch and produced a stable statewide exposure metric for the treatment variable.
4. Merging and Final Panel Construction:
   Datasets were harmonized and merged on a shared state–year index, with remaining gaps (mainly in smoking) imputed using interpolation and year-specific means. The result is a balanced panel containing outcomes, treatment, and confounders for every state–year, enabling valid regression and causal inference while acknowledging that imputation smooths some natural variation.

Step#2: Research Question

Research Question:

Does long-term exposure to high PM2.5 levels cause an increase in state-level hospitalization rates for asthma in the U.S.?

Research Significance:

Economic Cost-Benefit Analysis:
One real-world decision we could make by establishing the causal link is that governments could quantify the economic burden of PM2.5 through increased hospitalizations and healthcare costs. These estimates can guide investment decisions by showing the financial benefits of implementing pollution control policies.

Method Legitimacy:

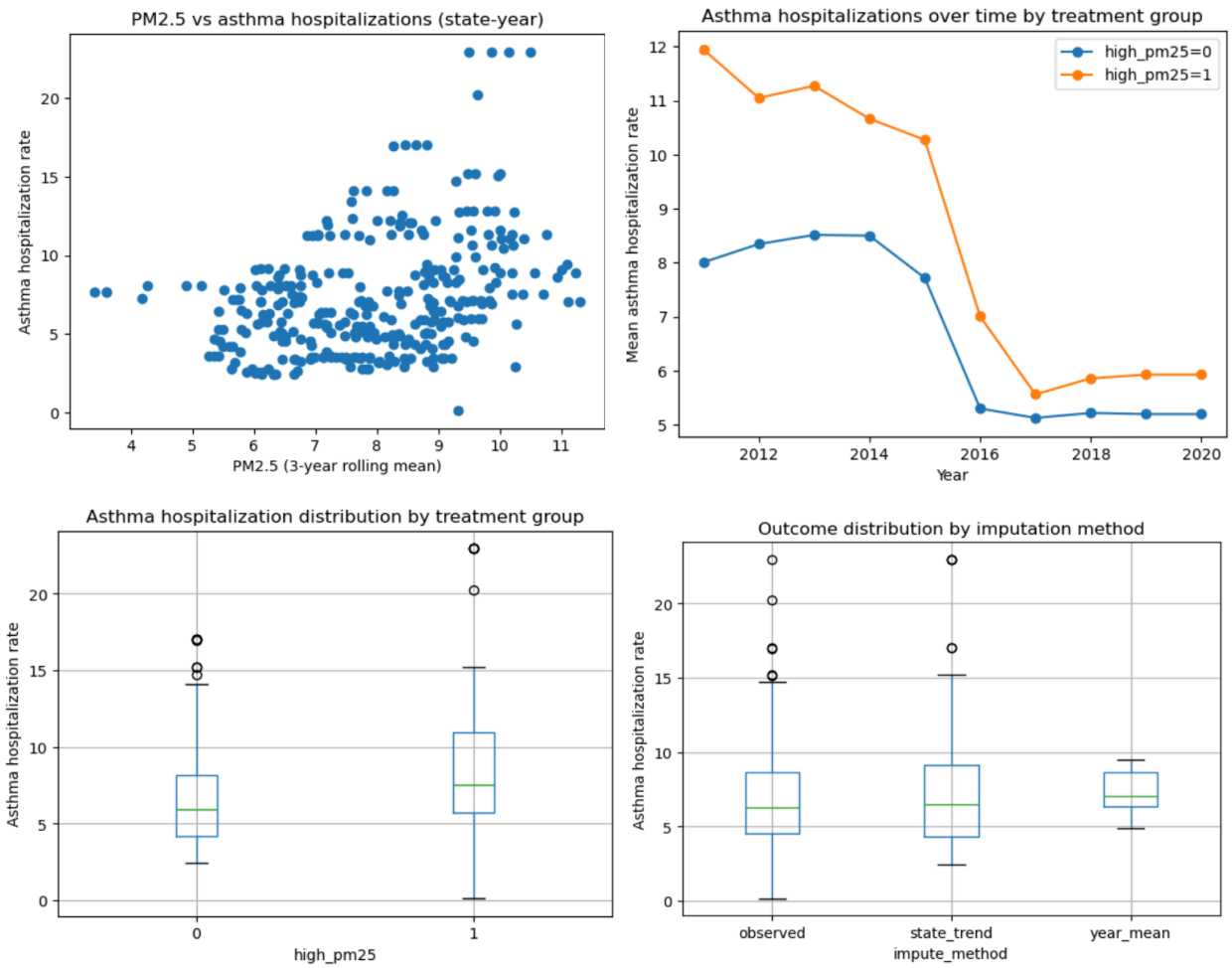
Causal inference is appropriate for our project because we can compare state-years with similar confounder profiles but different pollution levels, approximating the counterfactual contrast needed to estimate the effect of long-term PM2.5 exposure by using causal inference methods such as regression adjustment or propensity-score weighting

Method Limitations:

A key limitation of our approach is the assumption that all relevant confounders are observed and well measured; unobserved state-level factors—such as healthcare access, indoor air quality, or environmental policies—may still influence both PM2.5 and asthma hospitalizations. In addition, using aggregated state-level data introduces ecological bias, as averages cannot capture individual exposure or risk. Because PM2.5 variation across states is not random, residual structural differences may remain even after adjustment, limiting clean causal identification.

Step#3: EDA

Variables Visualization:



Trends/Relationship Observed:

The exploratory plots suggest a weak-to-moderate positive association between long-term PM2.5 and asthma hospitalization rates, with substantial variability and some outliers. High-PM states generally exhibit higher hospitalization rates and a heavier upper tail, while a sharp drop around 2016 in both groups indicates a strong year effect that should be controlled for. Boxplots also show that year-mean imputation reduces outcome variability, motivating sensitivity analyses comparing observed-only results to those including imputed outcomes.

Visualizations Relevance:

(Motivating the question? Suggesting a potential answer?)
These visualizations are directly tied to our causal question because they connect the exposure (PM2.5, including the long-term measure used in our treatment definition) to the outcome

(asthma hospitalizations) and show how the outcome differs across the treatment and control groups. The scatter plot and treatment boxplot motivate the question by providing initial evidence that higher pollution is associated with worse respiratory outcomes and suggest the direction of a potential effect. The time-series plot suggests why the causal model should include year fixed effects (shared national shifts) and why comparing within-state changes over time is important. The imputation diagnostic is relevant because it shows that data-processing choices can change the outcome distribution, so the final causal conclusions should be supported by robustness checks rather than a single specification.

Step#4: Method Introduced (Casual Inference)

Casual Inference Assumption:

The estimate is interpreted causally if (i) there are no important unobserved time-varying state factors that affect both PM2.5 and asthma hospitalizations after controlling for observed confounders and state/year fixed effects, (ii) each state-year could plausibly be in either exposure group (positivity), and (iii) one state's exposure does not change another state's outcome (SUTVA, approximately).

Experiment:

Randomly assign each state-year to "high long-term PM2.5" versus "not high," then compare asthma hospitalization rates.

1. **Treatment:**
   `high_pm25` = 1 (top-quartile PM2.5 in that year; long-term measure uses `pm25_mean_3yr`).
2. **Control:**
   `high_pm25` = 0.
3. **Units:**
   State-year observations (33 states, 2011–2020).

Method:

We estimate the causal effect of long-term PM2.5 exposure on asthma hospitalization rates using an observational causal inference framework at the state–year level. For each observation i, we define **asthma hospitalization rate Y_i as <u>outcome</u>, a <u>binary treatment</u> indicator Z_i equal to 1 for high long-term PM2.5 exposure and 0 otherwise, and a vector of observed pre-treatment <u>confounders</u> X_i, including smoking prevalence, obesity, physical inactivity, median household income, census region, and calendar year.**

Our estimand is the Average Treatment Effect (ATE):

$$\tau = \mathbb{E}[Y_i(1) - Y_i(0)]$$

where $Y_i(1)$ and $Y_i(0)$ denote potential outcomes under high and low PM2.5 exposure.

Because treatment assignment is not randomized, we adjust for confounding using propensity score subclassification. The propensity score is defined as

$$e(X_i) = \Pr(Z_i = 1 \mid X_i)$$

and is estimated via logistic regression:

$$\Pr(Z_i = 1 \mid X_i) = \text{logit}^{-1}\big(\beta_0 + X_i^\top \beta\big)$$

including region and year fixed effects.

We partition observations into subclasses with similar estimated propensity scores $\hat{e}_i$. Within each subclass k, we estimate the treated–control difference:

$$\hat{\tau}_k = \mathbb{E}[Y_i \mid Z_i = 1,\, i \in k] - \mathbb{E}[Y_i \mid Z_i = 0,\, i \in k]$$

and compute the overall ATE as a weighted average:

$$\hat{\tau} = \sum_k w_k \hat{\tau}_k$$

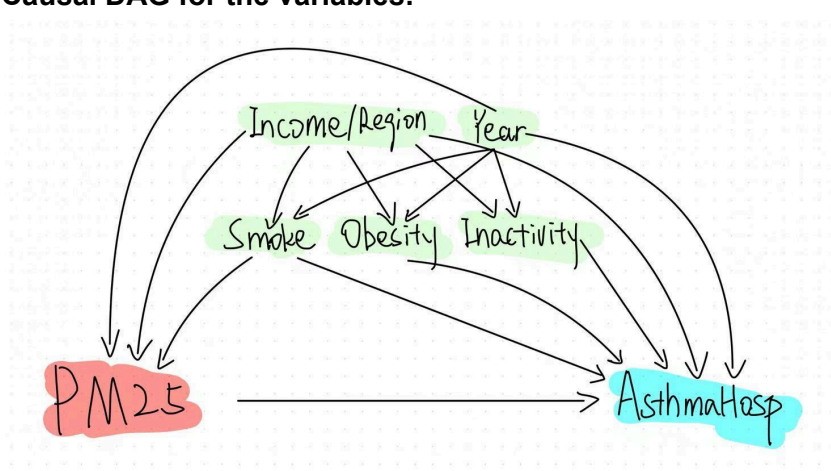
where w_k is the proportion of observations in subclass k.

Under the conditional independence assumption

$$(Y_i(0), Y_i(1)) \perp\!\!\!\perp Z_i \mid X_i$$

and assuming sufficient overlap, $\hat{\tau}$ can be interpreted as the causal effect of sustained PM2.5 exposure on asthma hospitalizations, conditional on observed confounders.

1. **Confoundedness or unconfoundedness? Why?**
   Confoundedness. High-PM state–years differ from low-PM state–years in observed behavioral, socioeconomic, and geographic characteristics that also affect asthma hospitalizations, motivating adjustment for these confounders in our analysis.
2. **Methods used to adjust for confounders:**
   Logistic propensity score model for high_pm25, trimming extreme scores (<0.05 or >0.90), then subclassification into 5 propensity-score groups and comparison of treated vs. control outcomes within each group. Balance tables check that confounder means are closer after subclassification.
3. **Any colliders in the dataset?**
   They are: Avoid conditioning on variables downstream of PM2.5 or healthcare demand, e.g., air-quality policy adoption, hospital capacity, or utilization, which can open back-door paths.
4. **Causal DAG for the variables:**



Interpretation:

- The DAG summarizes our identification assumptions at the state–year level.
- Long-term PM2.5 exposure is the treatment.
- Asthma hospitalizations are the outcome.

- Smoking is included as a direct confounder because it plausibly affects asthma risk and is correlated with pollution-intensive economic activity.
- Income, region, and calendar year capture persistent structural and temporal factors—such as industrial composition, regulation, and national trends—that jointly influence PM2.5 and asthma outcomes.
- Obesity and physical inactivity are treated as pre-treatment proxies for broader population health and socioeconomic environments rather than as direct causes of PM2.5. Conditioning on these variables aims to block backdoor paths driven by shared upstream factors.

We acknowledge ambiguity in some aggregate-level causal directions; thus, the DAG represents a working identification model used to justify conditional ignorability given observed covariates.

Results:

After subclassification, treated and controlled state-years within each propensity group have much closer means for smoking, obesity, and inactivity than in the raw data. The weighted average difference in asthma hospitalizations between high- and low-PM groups is about 0.8 extra hospitalizations per 10,000 in high-PM state-years.

```
high_pm25   0   1
ps_group
0          56   5
1          57   3
2          53   7
3          25  35
4          23  37
   group   n     tau_g
0      0  61 -1.165536
1      1  60  1.033684
2      2  60  0.985585
3      3  60  2.970246
4      4  60  0.137808
Subclassif. ATE (high_pm25 vs low): 0.7858528866943146
```

1. **Summarization of the results:**
   Signs are consistently positive; statistical significance weakens when restricting to observed outcomes.
2. **Causality statement:**
   Evidence is consistent with a harmful effect of long-term PM2.5 on asthma hospitalizations, but the size and significance are sensitive to outcome imputation.
3. **Magnitude of any effect:**
   Estimated ATE ≈ 0.8 additional hospitalizations per 10,000 residents for high-PM state-years compared with similar low-PM state-years.
4. **Uncertainty in the estimate/evidence against the hypotheses we're investigating:**
   Standard errors and a formal confidence interval for the subclassification ATE are not computed, but sample size is moderate (~330 state-years), and results are consistent with regression models including the same confounders. Some subclasses have few treated units, so estimates for those groups are noisy. Evidence against a null of zero effect is suggestive, not definitive.

Discussion (based on the results):

1. **Limitations of our methods:**
   (1) **Geographic coverage.** Our analysis includes only 33 states and one imputed PM2.5 value, limiting generalizability and adding uncertainty.
   (2) **Propensity-score.** Propensity-score subclassification improves balance but does not make treatment and control identical, especially in the highest-pscore group.
   (3) **Exposure measurement error.** Aggregating county PM2.5 to state averages obscures spatial heterogeneity and may misrepresent actual exposures.
   (4) **Unmeasured time-varying confounders.** Missing information on wildfire smoke, insurance coverage, and healthcare access may bias results if correlated with exposure and outcomes.
2. **Additional data that might be useful:**
   Wildfire smoke days, meteorology, insurance coverage, hospital capacity, age structure by state-year, poverty/education, race/ethnicity mix.
3. **Confidence about the casual relation and why:**
   Moderate; consistent across EDA, adjusted regressions, and the propensity-score ATE, and it matches external literature. Less confident about the exact size of the effect because of residual imbalance, potential unmeasured confounding, and limited sample size.
4. **Another plausible explanation for the effect observed in the data:**
   Differences in healthcare access/coding or contemporaneous policy changes correlated with PM2.5 could drive higher hospitalization rates independent of true prevalence.

Step#5: Prior Work

Prior Works Citation:

Guarnieri M., Balmes J.R. (2014). *Outdoor air pollution and asthma*. **The Lancet**, 383(9928), 1581–1592.
Open-access link: https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4465283/

Questions Relevance/Methods Comparison:

Relevance:
This review summarizes evidence that outdoor air pollutants, including PM2.5, trigger asthma exacerbations and increase hospitalizations—directly aligning with our question about whether higher PM2.5 levels raise asthma hospitalization rates.
Similarity:
Like our study, the authors rely on observational epidemiological evidence and focus on population-level asthma outcomes, emphasizing biological mechanisms that make PM2.5 harmful.
Differences:
Guarnieri and Balmes conduct a narrative synthesis of diverse epidemiological, clinical, and toxicological studies to establish biological plausibility and consistency of evidence, rather than estimating a single causal effect. In contrast, our study applies a propensity score–based causal inference framework to state–year panel data, explicitly modeling treatment assignment and estimating an average treatment effect under the conditional independence assumption. While their work establishes broad scientific consensus on the pollution–asthma link, our approach focuses on quantifying a population-level causal effect within a specific empirical setting, subject to explicit assumptions and limitations.

Results Comparison:

Our findings are consistent with Guarnieri and Balmes (2014), who report that exposure to outdoor air pollution, including PM2.5, is linked to increased asthma exacerbations and hospitalizations. Using a propensity score–based approach, we likewise find that state–years with high long-term PM2.5 exposure have higher asthma hospitalization rates than comparable low-exposure states–years after adjusting for observed confounders.

The studies differ in scope and methodology. Guarnieri and Balmes synthesize evidence from multiple epidemiological and clinical studies to establish broad scientific consensus, whereas our analysis estimates a single average treatment effect using U.S. state–year data and relies on explicit causal assumptions. Consequently, our results offer a more focused quantitative estimate, while theirs provide broader corroboration across diverse study designs.

Step#6: Conclusions

Outcome Summary/Key Findings:

- **Positive effect of PM2.5:** State–years with high long-term PM2.5 exposure exhibit higher asthma hospitalization rates than comparable low-exposure state–years after balancing observed confounders.
- **Causal evidence under assumptions:** Using propensity score subclassification, we estimate a positive average treatment effect, supporting a causal interpretation under the conditional independence and overlap assumptions.
- **Consistency across strata:** The treated–control differences are generally positive across propensity score strata, indicating that results are not driven by a small subset of observations.
- **Alignment with prior research:** Our findings are consistent with existing epidemiological evidence linking PM2.5 exposure to worsened asthma outcomes.
- **Limitations acknowledged:** The causal interpretation remains conditional on observed confounders, and unmeasured time-varying factors may still bias the estimates.

Critical Evaluation:

1. **Limitations in the data for our analysis:**
    - **Limited geographic coverage:**
      Hospitalization data are available for only 33 states, which restricts representativeness and may bias results if missing states differ systematically in pollution or asthma rates.
    - **Missing values requiring imputation.**
      Gaps in both the outcome and PM2.5 data required imputation, which smooths true variation and introduces uncertainty we cannot fully quantify.
    - **Lack of individual-level covariates.**
      State-level aggregates prevent us from controlling for demographic or socioeconomic factors that strongly influence asthma vulnerability and exposure.
2. **Helpful domain knowledge missing:**
   We lacked domain knowledge about which exposure time window—short-term spikes versus long-term cumulative PM2.5—is most relevant for asthma hospitalizations. In particular, it was unclear whether a three-year PM2.5 average aligns with clinical evidence. Input from an environmental health expert on the most predictive exposure window could materially affect our treatment definition and model design, helping better align our analysis with the biological mechanisms underlying asthma.

3. **Robustness of conclusions to our modeling choices:**
   Our conclusions are reasonably robust but depend on key modeling choices. In particular, defining high PM2.5 exposure using a relative threshold affects which state–years are treated and could change the estimated effect under alternative cutoffs. The results also depend on the specification of the propensity score model and the set of

confounders used to balance treated and untreated groups. Although subclassification improves covariate balance, unmeasured time-varying confounders may remain, so the estimated effect should be interpreted as causal conditional on these assumptions rather than as a definitive causal claim.

4. **Generalizability of our results:**
   Our findings are generalizable within a well-defined scope. Because the analysis is conducted at the U.S. state–year level and focuses on long-term PM2.5 exposure, the results primarily reflect population-level asthma outcomes in the United States during the study period. They may not extend to short-term pollution events, smaller geographic units, or settings with different environmental and healthcare contexts. Therefore, our conclusions should be viewed as context-specific rather than universally applicable.

Recommendations:

1. **Potential future studies that could build on our work:**
   One future study could extend our work by conducting a finer-grained causal analysis at the county or ZIP-code level using high-resolution PM2.5 and asthma outcome data. Our state-level estimates provide initial evidence that long-term PM2.5 increases asthma hospitalization risk and highlight where stronger identification is needed, particularly to capture local exposure variation and time-varying confounders. A follow-up study using higher-frequency pollution data and quasi-experimental designs (e.g., wildfire seasons or plant closures) could test whether similar causal effects persist at finer scales, helping distinguish true effects from aggregation bias.
2. **Call To Action and Potential Impacts of Our CTA:**
   Based on our causal estimates, states should consider adopting stricter PM2.5 pollution standards and enforcing stronger emission controls in persistently high-exposure regions. Our analysis suggests that residents of state–years with sustained high PM2.5 levels experience higher asthma hospitalization rates, even after adjusting for major behavioral, socioeconomic, and temporal confounders, indicating that current regulatory thresholds may not adequately protect respiratory health.

   State governments could tighten PM2.5 limits beyond federal minimums, prioritize mitigation near major sources such as industrial combustion and transportation corridors, and target counties that consistently fall into the high-exposure group. While these policies may increase compliance costs for pollution-intensive industries, the potential public health benefits are substantial. The largest gains would likely accrue to vulnerable populations—children, older adults, and low-income communities—who bear a disproportionate asthma burden. At the same time, stronger standards may encourage technological innovation and cleaner production methods, making PM2.5 reduction not only a public health imperative but also a cost-effective and forward-looking policy choice.

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Words Count (Before Reference Page): 3106